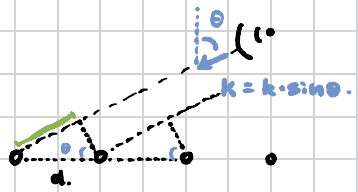
1. 定义与模型

→ For 1D scenario



$\Delta d = d\cdot\sin\theta$ $\qquad u = \sin\theta.$

$\Delta\Phi = 2\pi\cdot\frac{\Delta d}{\lambda} = 2\pi\cdot d\sin\theta/\lambda.$

针对 m 个阵元：阵元位置 $(m-1)\cdot d$.

阵列对应的 Array factor：$AF(u) = \sum_{m=0}^{N-1} \exp(jkmdu)$

几何级数.

$$\sum_{n=0}^{N-1} \gamma^n = \frac{1-\gamma^N}{1-\gamma}$$

$$AF(u) = \frac{1-\exp(jN\cdot kdu)}{1-\exp(j\cdot kdu)}$$

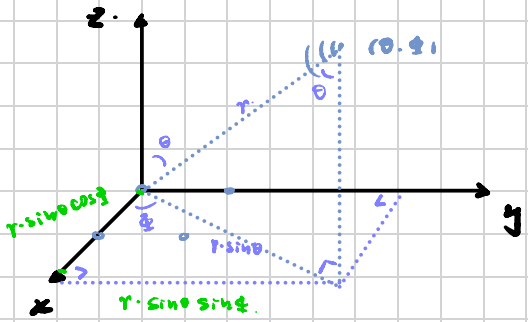
Assume:

$\Phi = N\cdot K\cdot d\cdot u.$

$$\sum_{n=0}^{N-1} e^{jn\Phi} = \exp(j(N-1)\Phi/2)\cdot\frac{\sin(N\Phi/2)}{\sin(\Phi/2)}$$

$$= \underbrace{\exp\left(j\frac{N-1}{2}\Phi\right)}_{\text{主瓣}}\ \underbrace{\frac{\sin N(\Phi/2)}{\sin(\Phi/2)}}_{\text{旁瓣}}$$

→ For 2D scenario



$$\begin{bmatrix} u = \sin\theta\cos\phi \\ v = \sin\theta\sin\phi \\ w = \cos\theta \end{bmatrix}$$

$$AF_x = \exp(jkdu(N-1)/2)\ \frac{\sin(N\,kdu/2)}{\sin(kdu/2)}$$

⇒ 让 main lobe 指向信号方向 ⟺ 补偿信号方向带来的阵列方向图偏转.

原先的扫描逻辑：AF assume 有入射波进行 Array Response 的操作.

有信号后的扫描：在真正有 incident 的方向叠加 J 相位差.

$(k\,m\,dx\cdot u_0,\ k\,m\,dy\cdot v_0)$ 步骤.

$$AF(u,v) = \sum_{m=0}^{N_x-1}\sum_{n=0}^{N_y-1} \exp(jk\,m\,dx(u-u_0) + n\,dy(v-v_0))$$